

External signals are converted to responses within the cell

Scientists think that signaling mechanisms first evolved hundreds of millions of years ago in ancient prokaryotes and single-celled eukaryotes and then were adapted for new uses in their multicellular descendants. So let's begin by considering signaling in some examples of single-celled organisms: bacteria and yeasts.

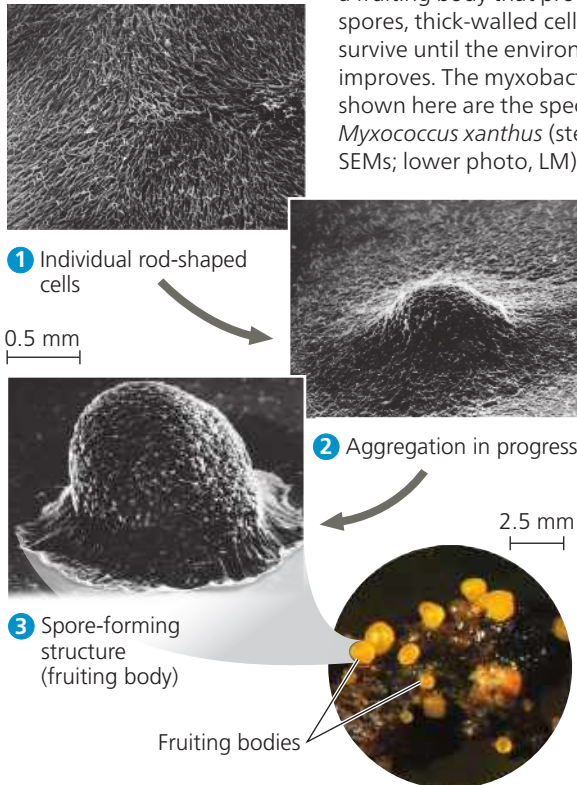
Evolution of Cell Signaling

EVOLUTION Research during the 1970s suggested that bacterial cells—somewhat surprisingly, since they are single-celled organisms—were capable of signaling to each other. Since then, we have come to understand that cell signaling is critical among prokaryotes. Bacterial cells secrete molecules that can be detected by other bacterial cells (Figure 11.2). Sensing the concentration of such signaling molecules allows bacteria to monitor their own local cell density, a phenomenon called *quorum sensing*.

Quorum sensing allows bacterial populations to coordinate the behavior of all cells in a population in activities that require a given density of cells acting at the same time. One example is formation of a *biofilm*, an aggregation of bacterial cells attached to a surface by molecules secreted by the cells, but only after the

▼ **Figure 11.2 Communication among bacteria.** Soil-dwelling bacteria called myxobacteria (“slime bacteria”) use chemical signals to share information about nutrient availability. When food is scarce, starving cells secrete a signaling molecule that stimulates neighboring cells to aggregate. The cells form a structure called

a fruiting body that produces spores, thick-walled cells that can survive until the environment improves. The myxobacteria shown here are the species *Myxococcus xanthus* (steps 1–3, SEMs; lower photo, LM).



cells have reached a certain density. The biofilm protects the cells in it, and they often derive nutrition from the surface they are on. Biofilms are believed to be involved in up to 80% of all human bacterial infections. You have probably encountered biofilms many times, perhaps without realizing it. The slimy coating on a fallen log or on leaves lying on a forest path, and even the film on your teeth each morning, are examples of bacterial biofilms. In fact, tooth-brushing and flossing disrupt biofilms that would otherwise cause cavities and gum disease.

Another example of bacterial behavior coordinated by quorum sensing is the secretion of toxins by infectious bacteria, which has serious medical implications. Sometimes treatment by antibiotics doesn't work with such infections because of antibiotic resistance that has evolved in a particular strain of bacteria. A promising alternative treatment would be to disrupt toxin production by interfering with the signaling pathways used in quorum sensing. In the **Problem-Solving Exercise**, you can participate in the process of scientific thinking involved in this novel approach.

Now let's look at an example of cell signaling in yeasts (single-celled fungi). Cells of the yeast *Saccharomyces cerevisiae*—which are used to make bread, wine, and beer—identify their sexual mates by chemical signaling when they reproduce sexually. There are two sexes, or mating types, called **a** and **α** (Figure 11.3). Each type secretes a specific

▼ **Figure 11.3 Communication between mating yeast cells.** *Saccharomyces cerevisiae* cells use chemical signaling to identify cells of the opposite mating type and initiate the mating process. The two mating types and their corresponding chemical signaling molecules, or mating factors, are called **a** and **α**.

